

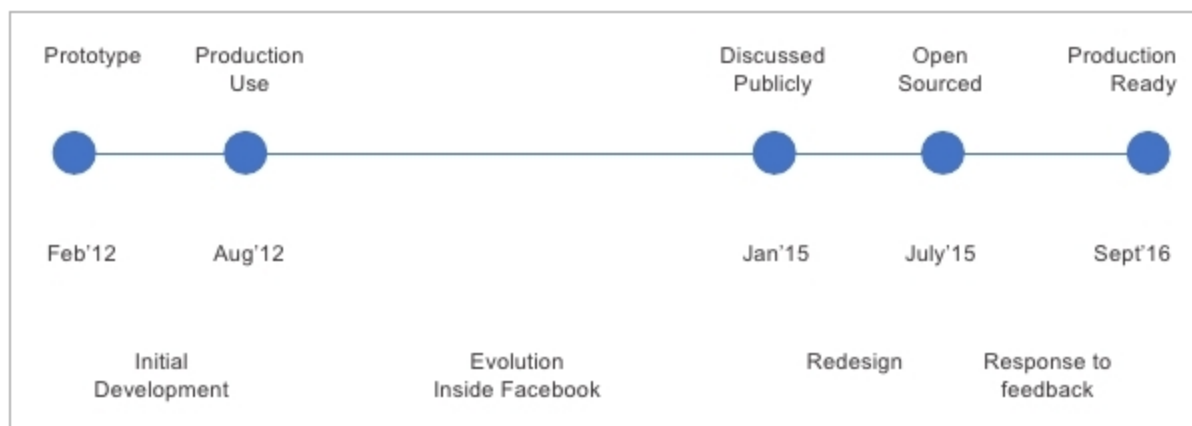


Figure 1: Evolution of GraphQL [Reference: Lessons from 4 Years of GraphQL, Lee Byron, GraphQL/Facebook, <https://www.youtube.com/watch?v=zVNrQo9XGOs&t=2156s>]

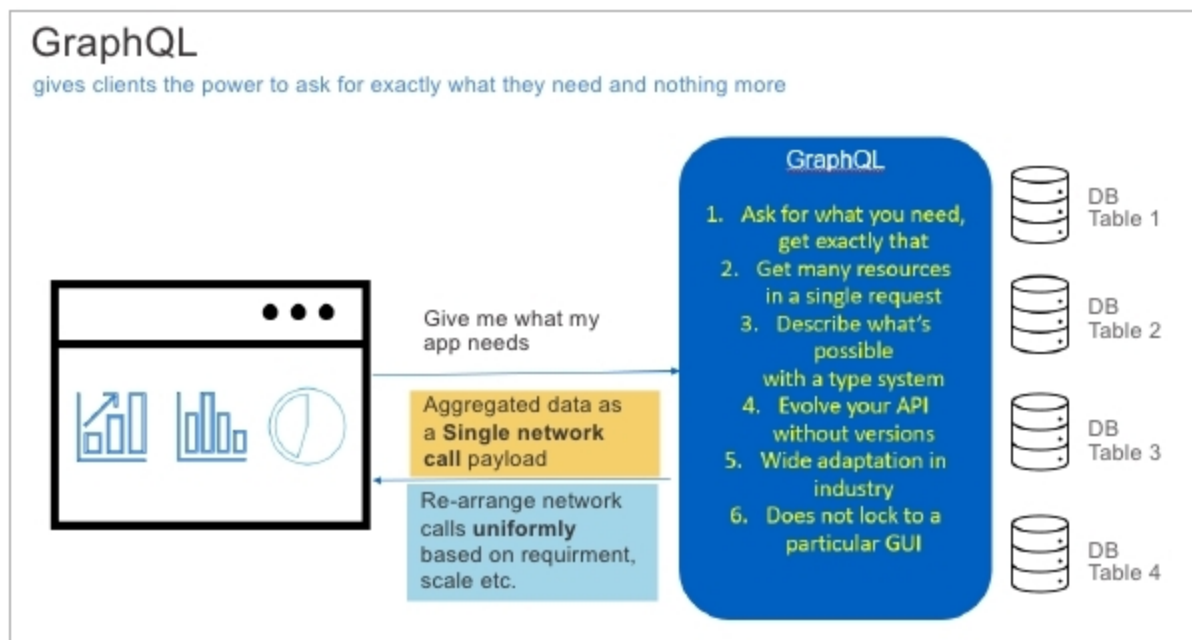


Figure 2: A helicopter view of GraphQL

1. Install Node.js from <https://nodejs.org/en/download/> and create a working directory for the first 'hello world' program:

```
λ node -v
v12.16.3
λ npm -v
6.4.1
λ mkdir my_app
λ cd my_app
```

2. Install VSCode in your working environment by downloading it from <https://code.visualstudio.com/download>.
3. Create a `package.json` in the project directory `my_app` (we are already inside `my_app`):

```
package.json
{
  "name": "graphql-hello-world",
```

```
  "private": true,
  "scripts": {
    "start": "nodemon --ignore data/ server.js"
  },
  "dependencies": {
    "notarealdb": "^0.2.2",
    "apollo-server-express":
    "^1.4.0",
    "body-parser": "^1.18.3",
    "cors": "^2.8.4",
    "express": "^4.16.3",
    "graphql": "^0.13.2",
    "graphql-tools": "^3.1.1"
  },
  "devDependencies": {
    "nodemon": "1.17.1"
  }
}
```

4. Install the dependencies through 'npm install'. This is going to take a few minutes:

```
λ npm install
```

5. Create a 'data' folder where we will create two files (our data store). Let's call the files `detectives.json` and `authors.json`:

`detectives.json`

```
[
  {
    "id": "D1",
    "name": "Prodosh C Mitter",
    "address": "21 Rajani Sen Road",
    "authorId": "A1"
  },
  {
    "id": "D2",
    "name": "Byomkesh Bakshi",
    "address": "Keyatala",
    "authorId": "A2"
  },
  {
    "id": "D3",
    "name": "Sherlock Homes",
    "address": "221B Baker Street",
    "authorId": "A3"
  },
  {
    "id": "D4",
    "name": "Hercule Poirot",
    "address": "Flat 203 at 56B
    Whitehaven Mansions",
    "authorId": "A4"
  },
  {
    "id": "D5",
    "name": "Miss Marple",
    "address": "Danemead Cottage",
    "authorId": "A4"
  }
]
```

`authors.json`

```
[
  {
    "id": "A1",
    "author": "Satyajit Ray",
    "assistant": "Topse"
  },
  {
    "id": "A2",
    "author": "Saradindoo
    Bandhyapadhyay",
```